



Advanced Machine Learning

2020/2021

Exam N 22 of June 2021 Time: 2h00

Name:

Number:

Question #	Points	Your score
1	18	
2	12	
3	20	
4	25	
5	25	
	100	

Grader:

Pergunta 1 - Define 18 points

Define the following concepts in a clear and **rigorous** way. You may use, if applicable, an example to **help you** to explain the concept.

a) Gradient Descent.

A sua resposta:

b) Overfitting.

A sua resposta:

c) Bayes Theorem.

A sua resposta:

Pergunta 2 - Comment 12 points

Comment the following statements. **Remember**, to comment is not just to say agree/disagree for you have to justify your evaluation.

- a) There is a difference between a **loss function** and a **quality metric**.

Your answer:

Your answer:

- b) When optimizing a **regression problem** where we know that only some of our features are useful, we should use **L1 regularization (Lasso)**.

Your answer:

Your answer:

Pergunta 3 - Solve 20 points

Consider the feature map and the filter/kernel of figure 1. What would be the resulting feature map after applying convolution with zero padding of 1 and a stride equal to 1?

Feature Map

1	0	0.5	0.5
0	0.5	1	0
0	1	0.5	1
1	0.5	0.5	1

Filter/Kernel

1	0
0	0.5

Figure 1: A feature map and a filter

Your answer:

Your answer:

Pergunta 4 - Explain/Describe 25 points

We discussed **discriminative** and **generative** approaches to learning. Explain what they have in common and also their differences. Do not forget to mention the types of problems that are most suited for each one.

Your answer:

Your answer:

Pergunta 5 - Architecture 25 points

We are witnessing the occurrence of the Euro2020 tournament. Suppose that for each particular game you want to **predict the winner**. Being a data scientist, explain your methodology from the beginning (data gathering and preparation) to the end (deployment of your system) and including the learning architecture. Justify carefully your choices.

Your answer:

Your answer:

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